

Self-consistent field theory of semiflexible nematics: Density–nematic coupling, anisotropic elasticity, and defect core sizes

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(Dated: June 30, 2026)

The linear response of wormlike chains in the nematic phase is studied by self-consistent field theory. The model Hamiltonian incorporates Maier–Saupe orientational interactions together with an isotropic excluded volume interaction. The latter models implicitly solvent mediated chain interactions, as appropriate for a lyotropic nematic. An effective free energy description for uniform nematic states is constructed in terms of the chain segment density and uniaxial nematic order parameter, providing a unified framework for density–degree of order coupling, isotropic–nematic coexistence, and the limit of stability of the nematic phase. Our results show that strong density–nematic degree of order coupling can destabilize the nematic state. The location of the instability depends on the ratio of excluded volume and nematic interaction, u_0/u_2 . In contrast, director distortions couple to density and nematic order variations only at higher order, remaining effectively decoupled in the linear response regime. The Frank elastic constants and the correlation lengths are obtained from a linear response analysis based on the self-consistent field theory free energy. Increasing flexibility strongly suppresses twist and bend elasticity while affecting splay elasticity comparatively weakly, leading to a crossover from bend-dominated to splay-dominated elasticity. The correlation lengths and Frank elastic anisotropy obtained from the linear response analysis explain well director profiles around a $+1/2$ disclination core, including the core size. The latter is proportional to the equilibrium correlation length, in agreement with Landau–de Gennes scaling.

I. INTRODUCTION

The linear response of small molecule, thermotropic liquid crystals is well understood [1–3]. On the other hand, comparable studies in lyotropic liquid crystals are more difficult, both theoretically and experimentally [4, 5]. Lyotropics comprise a very wide variety of molecular systems of differing architectures, including polymers, micellar and chromonic aggregates in solution, nanoparticle colloids, and bio filaments. Therefore, chain flexibility, hydrophobic and screened electrostatic interactions, and chain aggregation and scission all play a role in the response of the nematic phase depending on its molecular details. In particular, lyotropics generically exhibit large elastic anisotropy [6–9], the origin of which is not sufficiently understood. Despite the important structural differences between the two classes of liquid crystals, and by reason of symmetry, continuum descriptions of lyotropic liquid crystals are based on the same phenomenological Oseen–Frank and Landau–de Gennes theories that were originally developed for small molecule thermotropics. In fact, the Landau–de Gennes theory has become the de facto standard in studies of active nematic matter, including biopolymers [10–13].

Existing microscopic theories of liquid-crystalline ordering are largely based on the Onsager model and its extensions to semiflexible rods and polymers [14–17]. Although these theories have provided valuable insight into nematic ordering and phase behavior, the microscopic treatment of elasticity, defects, and density–nematic order coupling in spatially varying semiflexible nematics re-

mains challenging. Since these phenomena are strongly influenced by molecular flexibility, wormlike chain models within self-consistent field theory (SCFT) provide a natural theoretical framework [18]. Such a microscopic framework has been widely applied to the description of order in confined geometries [19], nematic–isotropic interfaces [20, 21], the determination of elastic constants [22, 23], and structure of topological defects [20]. We focus here solely on the effects of chain flexibility and osmotic compressibility on the linear response of a nematic phase by treating it as a wormlike chain. Our work follows earlier research on the subject [19, 20, 22–25]. The same formalism can be used to describe aggregation and scission of semiflexible aggregates [26, 27], and we will address them in future work. In addition, although very interesting phenomenology is known to arise from the interaction between screened electrostatics and chain flexibility [28–32], such an analysis is also beyond the scope of the present work. We mention that lyotropic chromonics and many biopolymers are comprised of weakly charged chains, and that such effects may be important to their linear response.

At a macroscopic level, the Landau–de Gennes theory has been widely used in the description of lyotropic liquid crystals, including details of their topological defects (see, e.g. Refs. [8, 12]), and a variety of nematic textures. The phenomenological coefficients appearing in the theory, including coefficients of gradient or elastic terms, are introduced without explicit connection to underlying molecular statistics. Hence, the microscopic origin of the large elastic anisotropy, for example, remains largely unexplored. The connection between the macroscopic free energy and chain statistics becomes particularly important in semiflexible lyotropic liquid crystals in which

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chain density (related to concentration in solution-based systems) and nematic order are strongly coupled, with chain flexibility significantly influencing elastic response. The statistical mechanical description presented here is capable of simultaneously capturing molecular flexibility, density fluctuations, and their coupling. In the case of lyotropic chromonic aggregates and other self-assembled liquid crystalline systems, the interplay between nematic order and aggregate formation and scission further complicates the connection between microscopic interactions and macroscopic continuum response. The analysis of this topic will be presented elsewhere [27].

We present the results of a SCFT calculation of the free energy of wormlike chains with nematic and excluded volume interactions, and a linear response analysis to investigate chain density–nematic order coupling, and the resulting elastic response. Elastic constants are shown to be anisotropic, and to include significant corrections for a compressible nematic that allows for chain density fluctuations. These results are interpreted in terms of an effective free energy, function of both chain density and nematic order parameter. Details of correlation lengths, Frank elastic constants, and topological defect profiles are analyzed numerically within the SCFT framework.

The remainder of this paper is organized as follows. In Sec. II, we present the SCFT framework for semi-flexible nematics based on a wormlike chain model with isotropic excluded volume and anisotropic Maier–Saupe interactions. In Sec. III, we construct an effective free energy description for uniform nematic states in terms of chain segment density and nematic order parameter, clarify the microscopic origin of density–nematic order coupling in the chain model, and determine the corresponding isotropic–nematic coexistence and spinodal boundaries. In Sec. IV, we introduce the linear response description of the nematic within the SCFT framework, and the calculation of Frank elastic constants, as well as density–nematic order variations around uniform states. Section V presents numerical results for density–nematic order coupling, and Frank elastic constants. The resulting correlation lengths associated with density and nematic order variations are analyzed and compared with defect core sizes. In addition, the relation between elastic anisotropy and director configurations around topological defects is investigated. Finally, conclusions and outlook are given in Sec. VI.

II. SCFT OF WORMLIKE CHAINS

We briefly summarize the SCFT framework for semi-flexible nematic polymers modeled as wormlike chains with Maier–Saupe (MS) interactions. This follows earlier theoretical developments [18, 19, 24, 25], and our previous work [20]. In addition to microscopic interactions, appropriate external fields are introduced to allow a linear response analysis.

We consider a system of n wormlike chains of contour

length L and persistence length l_p in a volume V at temperature T . The i -th chain configuration is described by a space curve $\mathbf{r}_i(s)$ with $s \in [0, L]$, and the unit tangent vector $\mathbf{u}_i(s) = d\mathbf{r}_i/ds$. The microscopic density of segment positions and orientations is defined as

$$\hat{\rho}(\mathbf{r}, \mathbf{u}) = \sum_{i=1}^n \int_0^L ds \delta(\mathbf{r} - \mathbf{r}_i(s)) \delta(\mathbf{u} - \mathbf{u}_i(s)), \quad (1)$$

which satisfies the normalization condition $\int d\mathbf{r} d\mathbf{u} \hat{\rho}(\mathbf{r}, \mathbf{u}) = nL$. The microscopic density of segments is given by $\hat{\rho}(\mathbf{r}) \equiv \int d\mathbf{u} \hat{\rho}(\mathbf{r}, \mathbf{u})$, which measures the local contour length per unit volume. For polymer melts, $\hat{\rho}(\mathbf{r})$ is naturally interpreted as a density field, whereas in lyotropic solutions it is more naturally interpreted as a concentration field, the two quantities differing only by an overall proportionality constant. The microscopic density $\hat{\rho}(\mathbf{r}, \mathbf{u})$ and $\hat{\rho}(\mathbf{r})$ are distinguished by their arguments. In some expressions, the arguments are suppressed for notational simplicity when the meaning is clear from the context. For a uniform state, the average density becomes spatially uniform, $\rho(\mathbf{r}) \equiv \langle \hat{\rho}(\mathbf{r}) \rangle = nL/V$. The Hamiltonian of the system without external fields is given by

$$\begin{aligned} \beta H_0 = & \frac{l_p}{2} \sum_{i=1}^n \int_0^L ds \left| \frac{d\mathbf{u}_i}{ds} \right|^2 \\ & + \frac{1}{2} \int d\mathbf{r} d\mathbf{r}' d\mathbf{u} d\mathbf{u}' \hat{\rho}(\mathbf{r}, \mathbf{u}) V_I(\mathbf{r}, \mathbf{r}'; \mathbf{u}, \mathbf{u}') \hat{\rho}(\mathbf{r}', \mathbf{u}'), \end{aligned} \quad (2)$$

where $\beta = 1/(k_B T)$, with k_B the Boltzmann constant. The first term represents the bending energy of individual chains, while the second term accounts for pairwise segment interactions. The interaction kernel is $V_I(\mathbf{r}, \mathbf{r}'; \mathbf{u}, \mathbf{u}') = \delta(\mathbf{r} - \mathbf{r}') [u_0 - u_2 P_2(\mathbf{u} \cdot \mathbf{u}')]$, where u_0 is the excluded volume parameter, u_2 is the MS interaction strength, and $P_2(x) = (3x^2 - 1)/2$ is the second order Legendre polynomial. In lyotropic and solution-based systems, solvent effects are treated implicitly through the isotropic excluded-volume interaction. In this interpretation, u_0 represents a solvent-renormalized excluded volume parameter whose value is related to the underlying Flory–Huggins interaction between chain and solvent particles [25, 33]. Its value depends on solution conditions such as solvent quality, salt concentration, and temperature, with good solvents generally corresponding to larger effective repulsion and poor solvents to weaker repulsion.

An external field is introduced via a linear coupling to $\hat{\rho}(\mathbf{r}, \mathbf{u})$,

$$\begin{aligned} \beta H_{\text{ext}} = & \sum_{i=1}^n \int_0^L ds v(\mathbf{r}_i(s), \mathbf{u}_i(s)) \\ = & \int d\mathbf{r} d\mathbf{u} v(\mathbf{r}, \mathbf{u}) \hat{\rho}(\mathbf{r}, \mathbf{u}). \end{aligned} \quad (3)$$

The total Hamiltonian is then given by $\beta H = \beta H_0 +$

βH_{ext} . The partition function of the system can be expressed as

$$Z = \frac{1}{n!v_0^n} \prod_{i=1}^n \int D^* \mathbf{r}_i \exp(-\beta H), \quad (4)$$

where v_0 is a reference volume introduced to make the partition function dimensionless. The asterisks on the path integral indicates that the integral is performed under the constraint $\prod_s \delta\left(\mathbf{u}(s) - \frac{d\mathbf{r}(s)}{ds}\right) \delta(|\mathbf{u}(s)| - 1)$. These delta function constraints enforce local inextensibility and unit tangent vectors along the chain contour. Applying a Hubbard–Stratonovich transformation to decouple the interaction term, the partition function can be rewritten, up to an irrelevant normalization constant, as a functional integral over a density field $\rho(\mathbf{r}, \mathbf{u})$ (introduced via the identity $\int \mathcal{D}\rho \delta[\rho - \hat{\rho}] = 1$, which enforces the constraint $\rho = \hat{\rho}$) and an auxiliary field $w(\mathbf{r}, \mathbf{u})$ conjugate to $\rho(\mathbf{r}, \mathbf{u})$,

$$Z \propto \int \mathcal{D}\rho \mathcal{D}w \exp(-\beta F[\rho, w]). \quad (5)$$

In the thermodynamic limit, the functional integral is evaluated using a saddle-point approximation. The saddle-point solution $w(\mathbf{r}, \mathbf{u})$ generally lies off the real axis. Following standard practice, a contour deformation (a rotation $w \rightarrow -iw$ in the complex plane) is performed such that $w(\mathbf{r}, \mathbf{u})$ becomes real. The corresponding free energy functional is written, up to additive constants and terms linear in n that only shift the reference chemical potential, as

$$\begin{aligned} \beta F[\rho, w] = & - \int d\mathbf{r} d\mathbf{u} [w(\mathbf{r}, \mathbf{u}) - v(\mathbf{r}, \mathbf{u})] \rho(\mathbf{r}, \mathbf{u}) \\ & - n \ln Z_1[w] + n \ln\left(\frac{nv_0}{V}\right) \\ & + \frac{1}{2} \int d\mathbf{r} d\mathbf{r}' d\mathbf{u} d\mathbf{u}' \rho(\mathbf{r}, \mathbf{u}) V_I(\mathbf{r}, \mathbf{r}'; \mathbf{u}, \mathbf{u}') \rho(\mathbf{r}', \mathbf{u}'). \end{aligned} \quad (6)$$

Here, $Z_1[w]$ denotes the single chain partition function, normalized by its value in the absence of external fields. In the saddle-point approximation, the free energy is evaluated at its stationary configurations. The stationary condition that follows from minimization with respect to $\rho(\mathbf{r}, \mathbf{u})$ gives

$$w(\mathbf{r}, \mathbf{u}) = \int d\mathbf{r}' d\mathbf{u}' V_I(\mathbf{r}, \mathbf{r}'; \mathbf{u}, \mathbf{u}') \rho(\mathbf{r}', \mathbf{u}') + v(\mathbf{r}, \mathbf{u}), \quad (7)$$

while minimization with respect to $w(\mathbf{r}, \mathbf{u})$ yields

$$\rho(\mathbf{r}, \mathbf{u}) = -n \frac{\delta \ln Z_1[w]}{\delta w(\mathbf{r}, \mathbf{u})}. \quad (8)$$

Equation (8) requires evaluating the single chain partition function in the field $w(\mathbf{r}, \mathbf{u})$ [18, 25],

$$\rho(\mathbf{r}, \mathbf{u}) = \frac{n}{4\pi V Z_1} \int_0^L ds q(\mathbf{r}, -\mathbf{u}, L-s; [w]) q(\mathbf{r}, \mathbf{u}, s; [w]), \quad (9)$$

where the single chain propagator $q(\mathbf{r}, \mathbf{u}, s; [w])$ satisfies the modified diffusion equation (MDE)

$$\frac{\partial q}{\partial s} = \left(\frac{1}{2l_p} \nabla_{\mathbf{u}}^2 - \mathbf{u} \cdot \nabla_{\mathbf{r}} - w(\mathbf{r}, \mathbf{u}) \right) q, \quad (10)$$

with initial condition $q(\mathbf{r}, \mathbf{u}, 0; [w]) = 1$. The single chain partition function is given by

$$Z_1[w] = \frac{1}{4\pi V} \int d\mathbf{r} d\mathbf{u} q(\mathbf{r}, \mathbf{u}, L; [w]). \quad (11)$$

At equilibrium, the saddle-point fields $\rho(\mathbf{r}, \mathbf{u})$ and $w(\mathbf{r}, \mathbf{u})$ must simultaneously satisfy the self-consistency equations, Eqs. (7) and (8). In practice, these equations are solved iteratively until convergence is achieved.

From the equilibrium distribution $\rho(\mathbf{r}, \mathbf{u})$, the orientational order parameter tensor can be computed [18],

$$\mathbf{Q}(\mathbf{r}) = \int d\mathbf{u} \rho(\mathbf{r}, \mathbf{u}) \left(\mathbf{u} \otimes \mathbf{u} - \frac{1}{3} \mathbf{I} \right), \quad (12)$$

where $\mathbf{I}$ is the identity tensor. $\mathbf{Q}(\mathbf{r})$ can be further parametrized as

$$\mathbf{Q}(\mathbf{r}) = \rho(\mathbf{r}) \left[S(\mathbf{r}) \left(\mathbf{n} \otimes \mathbf{n} - \frac{1}{3} \mathbf{I} \right) + P(\mathbf{r}) (\mathbf{m} \otimes \mathbf{m} - \mathbf{l} \otimes \mathbf{l}) \right], \quad (13)$$

where $\mathbf{n}(\mathbf{r})$, $\mathbf{m}(\mathbf{r})$, and $\mathbf{l}(\mathbf{r})$ define a local orthonormal triad, with $\mathbf{n}(\mathbf{r})$ denoting the local nematic director. The scalar fields $S(\mathbf{r})$ and $P(\mathbf{r})$ denote the uniaxial order parameter and the degree of biaxiality, respectively.

III. THE UNIFORM STATE

Within the SCFT framework, the equilibrium state is determined by the saddle-point equations (7) and (8), for $\rho(\mathbf{r}, \mathbf{u})$ and $w(\mathbf{r}, \mathbf{u})$. While this formulation provides a complete mean field description of the equilibrium state, the form of the free energy is not transparent. We first eliminate the auxiliary field $w(\mathbf{r}, \mathbf{u})$ by inverting Eq. (8) in the absence of external fields, $v(\mathbf{r}, \mathbf{u}) = 0$. Substituting the stationary solution $w^*[\rho(\mathbf{r}, \mathbf{u})]$ back into the free energy functional yields an effective free energy functional depending only on $\rho(\mathbf{r}, \mathbf{u})$. We then coarse grain the theory by expressing the free energy in terms of the low-order moments of $\rho(\mathbf{r}, \mathbf{u})$, namely $\rho(\mathbf{r})$ and $Q_{ij}(\mathbf{r})$. Finally, restricting the theory to uniform uniaxial states reduces the free energy to a function of the ρ and S .

The MS interaction contribution in Eq. (6) to the free energy is completely determined by the coarse grained fields $\rho(\mathbf{r})$ and $Q_{ij}(\mathbf{r})$, $\frac{1}{2} \int d\mathbf{r} (u_0 \rho^2 - \frac{3}{2} u_2 Q_{ij} Q_{ij})$, where repeated tensor indices imply summation throughout the paper. The relation in Eq. (7) shows that at a self-consistent solution the interaction generated auxiliary field has the same angular structure as the MS kernel, namely isotropic and quadrupolar components. Thus, we restrict the $w(\mathbf{r}, \mathbf{u})$ to this subspace,

$$w(\mathbf{r}, \mathbf{u}) = w_0(\mathbf{r}) + w_{ij}(\mathbf{r}) \left(u_i u_j - \frac{1}{3} \delta_{ij} \right). \quad (14)$$

Equation (8) becomes

$$\rho(\mathbf{r}) = -n \frac{\delta \ln Z_1[w_0, w_{ij}]}{\delta w_0(\mathbf{r})}, \quad (15)$$

$$Q_{ij}(\mathbf{r}) = -n \frac{\delta \ln Z_1[w_0, w_{ij}]}{\delta w_{ij}(\mathbf{r})}. \quad (16)$$

Inverting these relations yields the stationary fields $w_0^*[\rho, Q_{ij}]$ and $w_{ij}^*[\rho, Q_{ij}]$. Substituting them back into the free energy functional gives an effective free energy functional depending only on $\rho(\mathbf{r})$ and $Q_{ij}(\mathbf{r})$.

To obtain analytical insight into the functional form of the effective free energy, we restrict to a spatially uniform uniaxial state, where $\rho = nL/V$, $Q_{ij} = \rho S (n_i n_j - \frac{1}{3} \delta_{ij})$, and $w_{ij} = w_Q (n_i n_j - \frac{1}{3} \delta_{ij})$, with w_Q the scalar amplitude of the quadrupolar auxiliary field. The auxiliary field therefore becomes

$$w(\mathbf{u}) = w_0 + w_Q \left[(\mathbf{u} \cdot \mathbf{n})^2 - \frac{1}{3} \right]. \quad (17)$$

For the convention adopted here, the usual prolate nematic ordering ($S > 0$) corresponds to $w_Q < 0$, while oblate ordering ($S < 0$) corresponds to $w_Q > 0$. Equation (16) reduces to the scalar relation $S = -(3/2L) \partial \ln Z_1 / \partial w_Q$, which implicitly defines $w_Q^*(S)$.

For a spatially uniform state, the translational gradient term in the MDE, Eq. (10), vanishes. Factoring out the w_0 contribution through $q(\mathbf{u}, s) = e^{-w_0 s} \tilde{q}(\mathbf{u}, s)$, $\tilde{q}(\mathbf{u}, s)$ satisfies

$$\frac{\partial \tilde{q}(\mathbf{u}, s)}{\partial s} = \left[\frac{1}{2l_p} \nabla_{\mathbf{u}}^2 - w_Q \left((\mathbf{u} \cdot \mathbf{n})^2 - \frac{1}{3} \right) \right] \tilde{q}(\mathbf{u}, s), \quad (18)$$

with $\tilde{q}(\mathbf{u}, 0) = 1$. Therefore, the single chain partition function is $Z_1(w_0, w_Q) = e^{-w_0 L} Z_Q(w_Q)$, where Z_Q denotes the w_Q -dependent part. The free energy contribution of $e^{-w_0 L}$ cancels exactly with the term involving $w_0^* \rho$ in the first term of Eq. (6). Using the relation $w_Q^*(S)$, w_Q^* can be eliminated in favor of S , thereby expressing the effective free energy density in terms of ρ and S ,

$$\beta f_{\text{eff}}(\rho, S) = \frac{\rho}{L} \ln \left(\frac{\rho v_0}{L} \right) + \frac{1}{2} u_0 \rho^2 + \frac{\rho}{L} \left[g(S; L/l_p) - \frac{\alpha}{2} S^2 \right], \quad (19)$$

where $\alpha = u_2 \rho L = n L^2 u_2 / V$ can be interpreted as the effective strength of the alignment interaction. Here,

$$g(S; L/l_p) = -\frac{2}{3} \tilde{w}_Q^* S - \ln Z_Q(\tilde{w}_Q^*; L/l_p), \quad (20)$$

with $\tilde{w}_Q = L w_Q$. The function $g(S; L/l_p)$ contains the change in conformational entropy per chain associated with nematic ordering, measured relative to the isotropic reference state ($g(S=0) = 0$). The effective free energy form shows that ρ and S are coupled in two ways. First, the prefactor $\rho/L = n/V$ indicates that the free energy density is a sum over single chain contributions. Second, ρ modifies the shape of the S -dependent free energy landscape through the effective alignment strength

α . Minimization of the effective free energy with respect to S at constant ρ gives

$$g'(S; L/l_p) = \alpha S, \quad (21)$$

where the prime denotes differentiation with respect to S . From Eq. (20), one has $g'(S; L/l_p) = -\frac{2}{3} \tilde{w}_Q^*(S)$. Therefore, the extrema correspond to the intersections between the curves αS and $-\frac{2}{3} \tilde{w}_Q^*(S)$.

The chemical potential and osmotic pressure follow from the effective free energy density in Eq. (20) as

$$\begin{aligned} \beta \mu(\rho, S) &= L \frac{\partial(\beta f_{\text{eff}})}{\partial \rho} \\ &= \ln \left(\frac{\rho v_0}{L} \right) + 1 + u_0 \rho L + g - u_2 \rho L S^2, \\ \beta P(\rho, S) &= \rho \frac{\partial(\beta f_{\text{eff}})}{\partial \rho} - \beta f_{\text{eff}} \\ &= \frac{\rho}{L} + \frac{1}{2} u_0 \rho^2 - \frac{1}{2} u_2 \rho^2 S^2. \end{aligned} \quad (22)$$

The isotropic-nematic coexistence condition is obtained by requiring the chemical potential and osmotic pressure to be equal in the two uniform phases. Denoting the isotropic and nematic coexistence densities by ρ_I and ρ_N , respectively, we have $S = 0$ and $\alpha_I = u_2 \rho_I L$ in the isotropic phase, while $S = S_N$ and $\alpha_N = u_2 \rho_N L$ in the nematic phase. The nematic order parameter S_N is determined by Eq. (21) and corresponds to the stable solution with $S_N > 0$. The coexistence conditions are therefore

$$\begin{aligned} \ln \left(\frac{\alpha_N}{\alpha_I} \right) + g(S_N; L/l_p) + \frac{u_0}{u_2} (\alpha_N - \alpha_I) - \alpha_N S_N^2 &= 0, \\ \alpha_N - \alpha_I + \frac{1}{2} \frac{u_0}{u_2} (\alpha_N^2 - \alpha_I^2) - \frac{1}{2} \alpha_N^2 S_N^2 &= 0. \end{aligned} \quad (23)$$

For a given L/l_p , the coexistence conditions depend only on the single parameter u_0/u_2 . The resulting coexistence values α_I and α_N then yield the coexistence densities through $\rho_{I,N} = \alpha_{I,N}/(u_2 L)$.

The spinodal marks the limit of metastability of a uniform state. Define the Hessian matrix

$$A_{ij} \equiv \frac{\partial^2(\beta f_{\text{eff}})}{\partial x_i \partial x_j}, \quad x_{i,j} \in \{\rho, S\}, \quad (24)$$

evaluated at a uniform stationary state. The spinodal is reached when the smallest eigenvalue vanishes. Since $A_{\rho\rho} > 0$ throughout the physically relevant parameter range, this condition is equivalent to the condition $\det A = 0$ for the present two-variable problem. The isotropic spinodal is given by

$$\alpha_{I,\text{sp}} = g''(S=0; L/l_p), \quad (25)$$

where g'' denotes the second derivative with respect to S . The nematic spinodal is determined by

$$\left(\frac{1}{\alpha_{N,\text{sp}}} + \frac{u_0}{u_2} - S_N^2 \right) [g''(S_N; L/l_p) - \alpha_{N,\text{sp}}] - \alpha_{N,\text{sp}} S_N^2 = 0. \quad (26)$$

The results show that, for fixed L/l_p , the isotropic spinodal occurs at a constant value. In contrast, the nematic spinodal varies with the ratio u_0/u_2 .

The coexistence and spinodal conditions admit an equivalent formulation in terms of the grand potential density $\beta\omega(\mu, S) = \beta\Omega(\mu, S)/V$, where $\Omega = Vf_{\text{eff}} - \mu n$ is the grand potential. At coexistence, the isotropic and nematic minima of $\beta\omega$ have equal values. The spinodal corresponds to the point at which one of these minima loses stability, signaled by a vanishing curvature with respect to S at fixed chemical potential μ . The equivalence to the Hessian-based stability criterion is shown in Appendix A.

For finite persistence length l_p , Eq. (18) does not admit a simple closed-form solution. Consequently, the function $g(S; L/l_p)$ must generally be obtained numerically. In practice, this can be achieved by expanding the propagator in spherical harmonics and truncating the expansion at finite angular order. A notable exception is the rigid-chain limit ($l_p \rightarrow \infty$), in which the rotational diffusion term vanishes and the MDE can be solved analytically. The propagator is then given by

$$\tilde{q}(\mathbf{u}, s) = \exp \left[-w_Q s \left((\mathbf{u} \cdot \mathbf{n})^2 - \frac{1}{3} \right) \right]. \quad (27)$$

Therefore,

$$Z_Q(w_Q) = \frac{\sqrt{\pi}}{2} e^{Lw_Q/3} \begin{cases} \frac{\text{erfi}(\sqrt{-Lw_Q})}{\sqrt{-Lw_Q}}, & w_Q < 0, \\ \frac{\text{erf}(\sqrt{Lw_Q})}{\sqrt{Lw_Q}}, & w_Q > 0. \end{cases} \quad (28)$$

The closed form expression for $Z_Q(w_Q)$ enables an explicit parametric construction of $g(S)$ in the rigid-chain limit, with the relation $w_Q^*(S)$ obtained by numerically inverting $S(w_Q)$. The resulting functions $g(S)$, $g'(S)$, and $g(S) - \frac{1}{2}\alpha S^2$ are shown in Fig. 1(a, b). The function $g(S)$ exhibits singular behavior near the physical bounds $S = 1$ and $S = -1/2$. The competition between orientational entropy and alignment interactions produces a nematic minimum at finite S for sufficiently large α . As shown in the inset of Fig. 1(b), $g'(S) \sim (1 - S)^{-1}$ as $S \rightarrow 1$, implying a logarithmic asymptotic behavior $g(S) \sim -\ln(1 - S)$. This reflects the rapid loss of orientational entropy as the system approaches perfect alignment. Figure 1(c) shows the phase diagram in the rigid-chain limit, including the isotropic-nematic coexistence line (binodals) and the nematic spinodal. The ratio u_0/u_2 strongly influences the width of the coexistence region: decreasing u_0/u_2 broadens the coexistence window, consistent with previous studies [20, 24]. In addition, the location of the nematic spinodal depends on u_0/u_2 . Figure 1(d) shows the grand potential density for $u_0/u_2 = 1$ at the coexistence point and at the nematic spinodal. At coexistence, the isotropic and nematic minima have equal grand potential density, corresponding to the binodal shown in Fig. 1(c). Likewise, the nematic spinodal

corresponds to the point at which the nematic minimum loses stability, characterized by a vanishing curvature, $\partial^2(\beta\omega)/\partial S^2 = 0$, at the stationary point S_N .

The effective free energy constructed in this section is a local free energy density for spatially uniform states. For spatially inhomogeneous configurations, additional gradient contributions are expected to arise naturally from the full modified diffusion equation, through both the $\mathbf{u} \cdot \nabla$ transport term and the spatial dependence of $w(\mathbf{r}, \mathbf{u})$. In what follows, such gradient effects are instead investigated in the long-wavelength regime through the linear response formalism developed in the next section.

IV. LINEAR RESPONSE

While the effective free energy derived in the previous section describes uniform states, spatially varying perturbations are investigated through a linear response approach within the SCFT framework around a uniform bulk state. This requires introducing weak external fields that couple linearly to the relevant observables. In the present work, we consider external fields that couple linearly to three quantities: the segment density $\rho(\mathbf{r})$, the density weighted uniaxial order parameter, and the local director.

As reflected in the effective free-energy form in Eq. (19), ρ and S are coupled, requiring a coupled linear response analysis. In contrast, for small director distortions around a uniform uniaxial state, the leading order variation of the tensor order parameter $\delta Q_{ij}(\mathbf{r})$ is proportional to the local orientation angle, while variations in $\rho(\mathbf{r})$ and $S(\mathbf{r})$ enter only at higher order. Director distortions are therefore effectively decoupled from $\rho(\mathbf{r})$ and $S(\mathbf{r})$ at linear order, allowing the director response to be analyzed independently.

A. Nematic elasticity

We first focus on the linear response of the director field in a uniaxial nematic phase. In the long-wavelength limit, the elastic free energy cost of director distortions is described by the Frank free energy [3]

$$F = \int d\mathbf{r} \left[\frac{1}{2} K_1 (\nabla \cdot \mathbf{n})^2 + \frac{1}{2} K_2 (\mathbf{n} \cdot \nabla \times \mathbf{n})^2 + \frac{1}{2} K_3 |\mathbf{n} \times (\nabla \times \mathbf{n})|^2 \right]. \quad (29)$$

Here, K_1 , K_2 , and K_3 are elastic constants corresponding to the splay, twist, and bend elastic modes, respectively. To isolate individual deformation modes, we consider one-dimensional director distortions along the x -direction and parameterize the director $\mathbf{n}(x)$ in terms of a single angle field $\theta(x)$, which represents deviations from a reference orientation $\mathbf{n}_0$. Different deformation modes (splay, twist, and bend) can be realized by appropriate

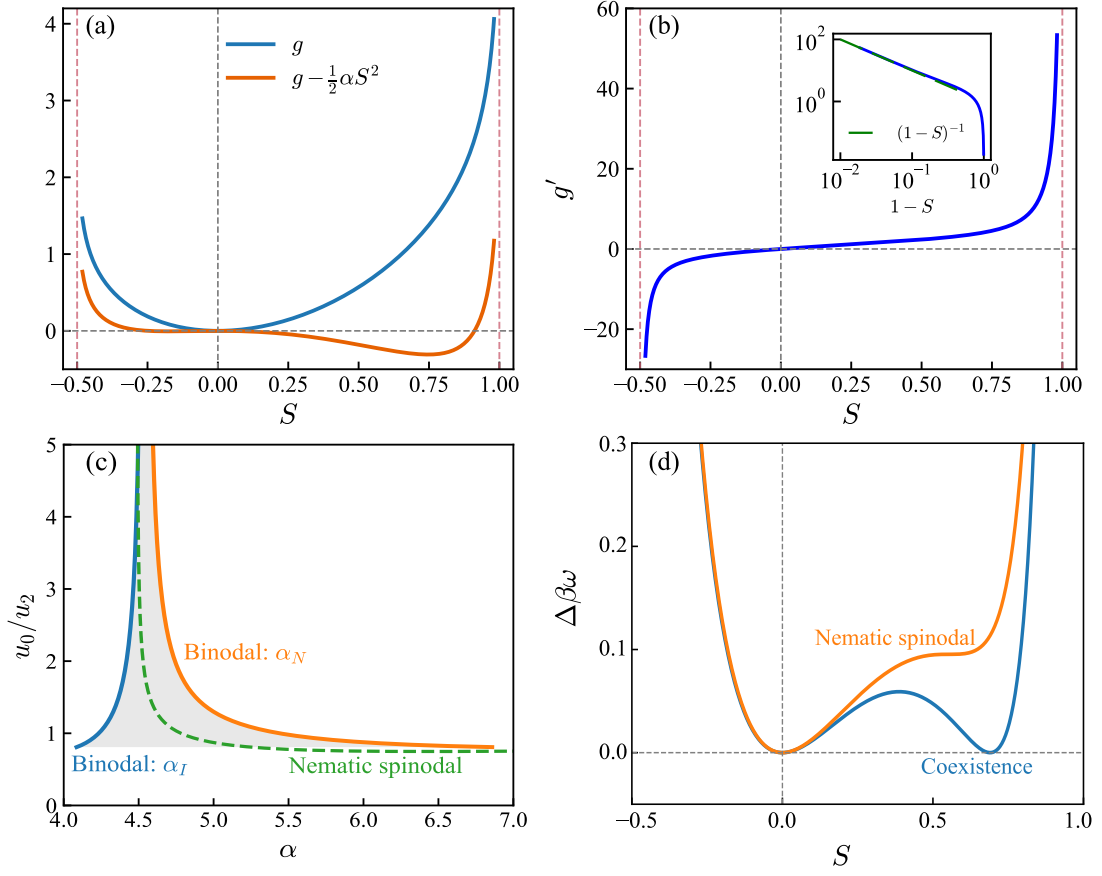


FIG. 1. (a) $g(S)$ and $g(S) - \frac{1}{2}\alpha S^2$ for $\alpha = 6$, and (b) $g'(S)$, both in the rigid-chain limit. The outer dashed vertical lines indicate the physical bounds $S = 1$ and $S = -1/2$. Inset of (b): log-log plot showing the asymptotic scaling of $g'(S)$ near the perfectly aligned limit. The green dashed line denotes the reference scaling $(1 - S)^{-1}$, showing that $g'(S) \sim (1 - S)^{-1}$ as $S \rightarrow 1$. (c) Phase diagram in the rigid-chain limit, showing the isotropic binodal (α_I), nematic binodal (α_N), and nematic spinodal ($\alpha_{N,sp}$) as functions of u_0/u_2 . (d) Grand potential difference $\Delta\beta\omega(S) = \beta\omega(S) - \beta\omega(0)$ for $u_0 = u_2 = 1$ and $L = 1$ in the rigid-chain limit. The two curves correspond to chemical potentials chosen at the coexistence point and at the nematic spinodal.

choices of the reference orientation $\mathbf{n}_0$ and the plane of director variation. Specifically, splay, twist, and bend correspond to director variations in the xz , yz , and xz planes, with the reference orientation $\mathbf{n}_0$ along the z , z , and x directions, respectively. In the small distortion limit, the Frank free energy density for all three deformation modes reduces to a common quadratic form,

$$f_i \simeq \frac{1}{2} K_i \left(\frac{d\theta}{dx} \right)^2, \quad i = 1, 2, 3. \quad (30)$$

As a representative example, we consider twist deformation, parameterized by $\mathbf{n} = (0, \sin \theta, \cos \theta)$ with $|\theta| \ll 1$. To excite a pure twist mode in linear response, we introduce an external field $v_\theta(\mathbf{r}, \mathbf{u}) = -h_\theta(x) u_y u_z$ that couples to the off-diagonal component Q_{yz} , with Hamiltonian

$$\beta H_{\text{ext}} = - \int d\mathbf{r} h_\theta(x) Q_{yz}(x). \quad (31)$$

For small distortions around a uniaxial nematic state,

the tensor field $\mathbf{Q}(x)$ is assumed to remain approximately uniaxial and can be written as $Q_{ij}(x) \simeq \rho(x) S(x) [n_i(x) n_j(x) - \frac{1}{3} \delta_{ij}]$. To leading order in the small angle, one obtains $Q_{yz}(x) \simeq \rho(x) S(x) \theta(x)$. Expanding about a uniform reference state and retaining only linear order terms in the distortions, we obtain $Q_{yz}(x) \simeq \rho_0 S_0 \theta(x)$, where ρ_0 and S_0 denote the density and uniaxial order parameter of the reference state. Variations in ρ and S contribute only at higher order and are therefore neglected at the linear response level. Accordingly, the external field reduces to an effective linear coupling to the rotation angle,

$$\beta H_{\text{ext}} \simeq - \int d\mathbf{r} \rho_0 S_0 h_\theta(x) \theta(x). \quad (32)$$

For one-dimensional distortions described by $\theta(x)$, the elastic free energy is proportional to $K_2 k^2 \theta(k)^2$ in Fourier space, while the external field provides a linear term proportional to $\rho_0 S_0 h_\theta(k) \theta(k)$. Here $k = 2\pi/\lambda$ denotes the magnitude of the wavevector, where λ is the corresponding distortion wavelength. Minimizing the free energy

yields

$$\theta(k) = \chi_2(k) h_\theta(k) = \frac{\rho_0 S_0}{\beta K_2 k^2} h_\theta(k), \quad (33)$$

where χ_2 is the linear susceptibility associated with the twist mode. This expression exhibits the characteristic k^{-2} scaling of a Goldstone mode. The same construction can be applied to the splay and bend modes by choosing the corresponding deformation geometry and probe fields, leading to mode dependent susceptibilities $\chi_i(k) = \rho_0 S_0 / (\beta K_i k^2)$ for $i = 1, 2, 3$. It thus provide a direct route to determine the Frank elastic constants from their long-wavelength response.

B. Density–nematic order coupling

We now turn to the linear response to density and uniaxial order parameter variations. For the density part, consider an orientation-independent (isotropic) external field that couples to the local segment density, $v_\rho(\mathbf{r}, \mathbf{u}) = -h_\rho(\mathbf{r})$. The corresponding energy is

$$\beta H_{\text{ext}} = - \int d\mathbf{r} h_\rho(\mathbf{r}) \hat{\rho}(\mathbf{r}), \quad (34)$$

where $\hat{\rho}(\mathbf{r}) = \int d\mathbf{u} \hat{\rho}(\mathbf{r}, \mathbf{u})$. For the uniaxial order parameter, consider a uniaxial field with easy axis $\mathbf{e}$, $v(\mathbf{r}, \mathbf{u}) = -h_Q(\mathbf{r}) [(\mathbf{u} \cdot \mathbf{e})^2 - \frac{1}{3}]$, where $h_Q(\mathbf{r})$ controls the alignment strength to the external field. The corresponding contribution to the free energy is given by

$$\beta H_{\text{ext}} = - \int d\mathbf{r} h_Q(\mathbf{r}) E_{ij}(\mathbf{e}) Q_{ij}(\mathbf{r}), \quad (35)$$

where $E_{ij}(\mathbf{e}) = e_i e_j - \frac{1}{3} \delta_{ij}$ is the traceless uniaxial tensor associated with $\mathbf{e}$. We define the corresponding scalar projection $Q(\mathbf{r}) \equiv Q_{ij}(\mathbf{r}) E_{ij}(\mathbf{e})$, so that $h_Q(\mathbf{r})$ couples linearly to $Q(\mathbf{r})$. Throughout the following analysis, the boldface symbol $\mathbf{Q}$ denotes the full nematic order tensor, while the non-bold symbol Q denotes the scalar quantity.

For the linear response analysis, we consider a spatially uniform reference state. In the nematic reference state, we choose the external field to be aligned with the director, $\mathbf{e} = \mathbf{n}$, so that the response probes density and uniaxial order parameter variations without mixing in director distortions. While in the isotropic reference state, the director $\mathbf{n}$ is induced by the external perturbation and coincides with $\mathbf{e}$. Assuming that the system remains locally uniaxial under small perturbations, we approximate $Q_{ij}(\mathbf{r}) \simeq \rho(\mathbf{r}) S(\mathbf{r}) (n_i n_j - \frac{1}{3} \delta_{ij})$. With this choice, the projection simplifies to $Q(\mathbf{r}) \simeq \frac{2}{3} \rho(\mathbf{r}) S(\mathbf{r})$, showing that Q represents a density-weighted nematic order parameter.

The response to small field induced deviations from the uniform reference state is analyzed in Fourier space. Introducing the wavevector $\mathbf{k}$, one defines

$$\delta \mathbf{x}(\mathbf{k}) = \begin{pmatrix} \delta \rho(\mathbf{k}) \\ \delta Q(\mathbf{k}) \end{pmatrix}, \quad \mathbf{h}(\mathbf{k}) = \begin{pmatrix} h_\rho(\mathbf{k}) \\ h_Q(\mathbf{k}) \end{pmatrix}, \quad (36)$$

where $\delta \mathbf{x}(\mathbf{k})$ represents ρ and Q deviations with respect to the reference state, and $\mathbf{h}(\mathbf{k})$ are the corresponding external fields. Within linear response theory, the free energy cost of small long-wavelength deviations from the uniform state can be expressed to quadratic order as

$$\beta F^{(2)} = \frac{1}{2} \sum_{\mathbf{k}} \delta \mathbf{x}^T(-\mathbf{k}) \mathbf{M}(\mathbf{k}) \delta \mathbf{x}(\mathbf{k}) - \sum_{\mathbf{k}} \mathbf{h}^T(-\mathbf{k}) \mathbf{x}(\mathbf{k}), \quad (37)$$

where $\mathbf{M}(\mathbf{k})$ is the inverse susceptibility kernel. The off-diagonal terms in M_{ij} quantify the coupling between ρ and Q . By construction, $\mathbf{M}(\mathbf{k})$ is symmetric, such that $M_{\rho Q} = M_{Q\rho}$. In the long-wavelength limit ($k \equiv |\mathbf{k}| \rightarrow 0$), $\mathbf{M}(\mathbf{k})$ is expanded to quadratic order in $\mathbf{k}$,

$$M_{ij}(\mathbf{k}) = A_{ij} + K_{ij}^{\alpha\beta} k_\alpha k_\beta, \quad (38)$$

where A_{ij} describes the uniform ($k = 0$) response and $K_{ij}^{\alpha\beta}$ characterize the leading response to spatial variations, with $i, j \in \{\rho, Q\}$ and $\alpha, \beta \in \{x, y, z\}$. The coefficients A_{ij} play the same role as the Hessian matrix introduced in Sec. III, representing the response to uniform variations. Here the variables are chosen as (ρ, Q) rather than (ρ, S) , so the individual matrix elements differ numerically. Nevertheless, both formulations describe the same underlying stability of the uniform state and therefore lead to the same spinodal condition, which, for the present two-variable problem, is equivalent to the vanishing of $\det A$, i.e., $A_{\rho\rho} A_{QQ} - A_{\rho Q}^2 = 0$. This condition shows that the onset of instability is controlled by the coupling between ρ and Q . In the absence of coupling ($A_{\rho Q} = 0$), the instability would occur independently in each channel when either $A_{\rho\rho} = 0$ or $A_{QQ} = 0$. A finite coupling modifies this criterion and can drive the system unstable even when both diagonal terms remain positive. According to Eq. (19), the resulting coefficients obtained after transforming variables from (ρ, S) to (ρ, Q) , with $S = 3Q/(2\rho)$, are

$$\begin{aligned} A_{\rho\rho} &= u_0 + \frac{1}{L\rho} + \frac{S^2}{L\rho} g'', \\ A_{\rho Q} &= -\frac{3S}{2L\rho} g'', \\ A_{QQ} &= -\frac{9}{4} u_2 + \frac{9}{4L\rho} g''. \end{aligned} \quad (39)$$

For a uniaxial reference state with director $\mathbf{n}_0$, rotational symmetry implies

$$K_{ij}^{\alpha\beta} = K_{ij}^\perp \delta_{\alpha\beta} + (K_{ij}^\parallel - K_{ij}^\perp) n_{0\alpha} n_{0\beta}, \quad (40)$$

so that

$$K_{ij}^{\alpha\beta} k_\alpha k_\beta = K_{ij}^\perp k_\perp^2 + K_{ij}^\parallel k_\parallel^2, \quad (41)$$

where $k_\parallel = \mathbf{k} \cdot \mathbf{n}_0$, $k_\perp^2 = k^2 - k_\parallel^2$. For brevity, labels indicating the perpendicular or parallel direction will be omitted when only one direction is under consideration.

The linear response follows from minimizing the free energy, yielding

$$\delta \mathbf{x}(\mathbf{k}) = \boldsymbol{\chi}(\mathbf{k}) \mathbf{h}(\mathbf{k}), \quad (42)$$

where $\boldsymbol{\chi}(\mathbf{k}) \equiv \mathbf{M}^{-1}(\mathbf{k})$ is the susceptibility matrix. The matrix elements of the susceptibility $\boldsymbol{\chi}(\mathbf{k})$ can be obtained from the linear response to weak external fields $h_\rho(\mathbf{k})$ and $h_Q(\mathbf{k})$ applied separately (with the other set to zero). The induced variations in ρ and Q define the response amplitudes, whose ratios to the applied field strength yield all four elements of $\boldsymbol{\chi}(\mathbf{k})$.

To gain further physical insight, we analyze the coupled response in terms of the eigenmodes of $\mathbf{M}(\mathbf{k})$. The corresponding eigenvalues are

$$\lambda_{\pm}(\mathbf{k}) = \frac{1}{2} \left[M_{\rho\rho} + M_{QQ} \pm \sqrt{(M_{\rho\rho} - M_{QQ})^2 + 4M_{\rho Q}^2} \right]. \quad (43)$$

The collective behavior is dominated by the mode with the smaller eigenvalue, $\lambda_{-}(\mathbf{k})$, referred to as the soft mode. Expanding the soft mode eigenvalue in the long-wavelength limit gives

$$\lambda_{-}(\mathbf{k}) \approx A_{-} + K_{-}^{\perp} k_{\perp}^2 + K_{-}^{\parallel} k_{\parallel}^2. \quad (44)$$

The associated longitudinal and transverse correlation lengths are defined as $\xi_{-}^{\parallel, \perp} = \sqrt{K_{-}^{\parallel, \perp} / A_{-}}$. The correlation length $\xi_{-}^{\parallel, \perp}$ characterizes the spatial scale of the soft mode, governing the decay of the coupled modes.

While the soft mode characterizes the collective responses of the coupled system, the quantity most directly accessed in simulations and experiments is Q . It is therefore useful to define a corresponding correlation length associated with this observable. We consider the projected response in the Q channel by minimizing with respect to ρ , which yields an effective susceptibility $\chi_Q(\mathbf{k}) = [\mathbf{M}^{-1}(\mathbf{k})]_{QQ}$. Correspondingly, its inverse takes the form

$$\chi_Q^{-1}(\mathbf{k}) = M_{QQ}(\mathbf{k}) - \frac{M_{\rho Q}^2(\mathbf{k})}{M_{\rho\rho}(\mathbf{k})}. \quad (45)$$

In the long-wavelength limit, it can be expanded as $\chi_Q^{-1}(\mathbf{k}) \approx A_Q + K_Q^{\perp} k_{\perp}^2 + K_Q^{\parallel} k_{\parallel}^2$, thereby defining effective coefficients A_Q and $K_Q^{\parallel, \perp}$. The corresponding correlation length is then given by $\xi_Q^{\parallel, \perp} \equiv \sqrt{K_Q^{\parallel, \perp} / A_Q}$.

V. RESULTS AND DISCUSSION

Before presenting our numerical results, we introduce a dimensionless formulation by rescaling all energy related and length related variables. All energy related quantities are made dimensionless using $\beta = 1/(k_B T)$, e.g., βF and βH . All quantities involving a length scale are rescaled by a reference length L_0 , such that $\tilde{l}_p = l_p / L_0$,

$\tilde{L} = L / L_0$, $\tilde{\mathbf{r}} = \mathbf{r} / L_0$ and $\tilde{u}_{0,2} = u_{0,2} / L_0$. In particular, the Frank elastic constants K_i are made dimensionless as $\beta \tilde{K}_i = \beta K_i L_0$. For notational simplicity, tildes are omitted in the following. Unless otherwise stated, all numerical results presented below are obtained from SCFT calculations using a real spherical harmonics representation. In practice, the expansion is truncated at finite angular order, which tends to underestimate the equilibrium nematic order parameter. Nevertheless, for the intermediate ordering regime, the truncation provides a reliable description. Details of the numerical implementation are provided in Appendix B.

We first summarize the isotropic-nematic transition at fixed density in this model, which has been investigated previously [20, 25]. The main results are briefly reviewed here to establish the notation and physical context for the following analysis. For a uniform state at fixed density, the equilibrium nematic order parameter is determined by the stationarity condition Eq. (21). A nematic solution exists only when $\alpha > \alpha^*(L/l_p)$, such that the nematic branch emerges above a threshold effective alignment strength. The corresponding nematic state is characterized by $S_N = S_N(\alpha, L/l_p)$. Within the nematic regime, numerical results show that S_N increases monotonically with α at fixed L/l_p , whereas at fixed α it decreases with increasing L/l_p .

A. Director response and Frank elastic constants

Frank elastic constants are obtained from the director response by varying the wavevector magnitude k and computing $\chi_i(k)$. The inverse susceptibility $\chi_i^{-1}(k)$ exhibits a linear dependence on k^2 within the range considered, with a negligible intercept, consistent with the expected Goldstone-mode behavior in Eq. (33). Numerical calculations further show that the induced variations in ρ and S do not contain a component at wavevector k , but instead appear only at $2k$. This confirms that, within linear response, the director variations are effectively decoupled from ρ and S . The numerical results suggest a scaling form for the elastic constants,

$$\beta K_i = \frac{1}{u_2} \tilde{K}_i(\alpha, L/l_p), \quad (46)$$

where $\tilde{K}_i$ is a scaling function. Data obtained for different values of L , u_2 , and n/V collapse onto this representation. In contrast, varying u_0 has negligible effect on βK_i , indicating that director elasticity is largely insensitive to compressibility.

Next, we examine the dependence of $\tilde{K}_i$ on α for several representative values of chain flexibility, $L/l_p = 0, 1, 5$, spanning the range from the rigid-chain limit to semiflexible chains. The results are shown in Fig. 2, corresponding to the nematic order range $S_N \sim 0.4$ – 0.7 . In the rigid-chain limit, $\tilde{K}_1$ and $\tilde{K}_2$ remain comparable, while $\tilde{K}_3$ increases more rapidly with α . In contrast, for semiflexible chains, both $\tilde{K}_1$ and $\tilde{K}_3$ become significantly

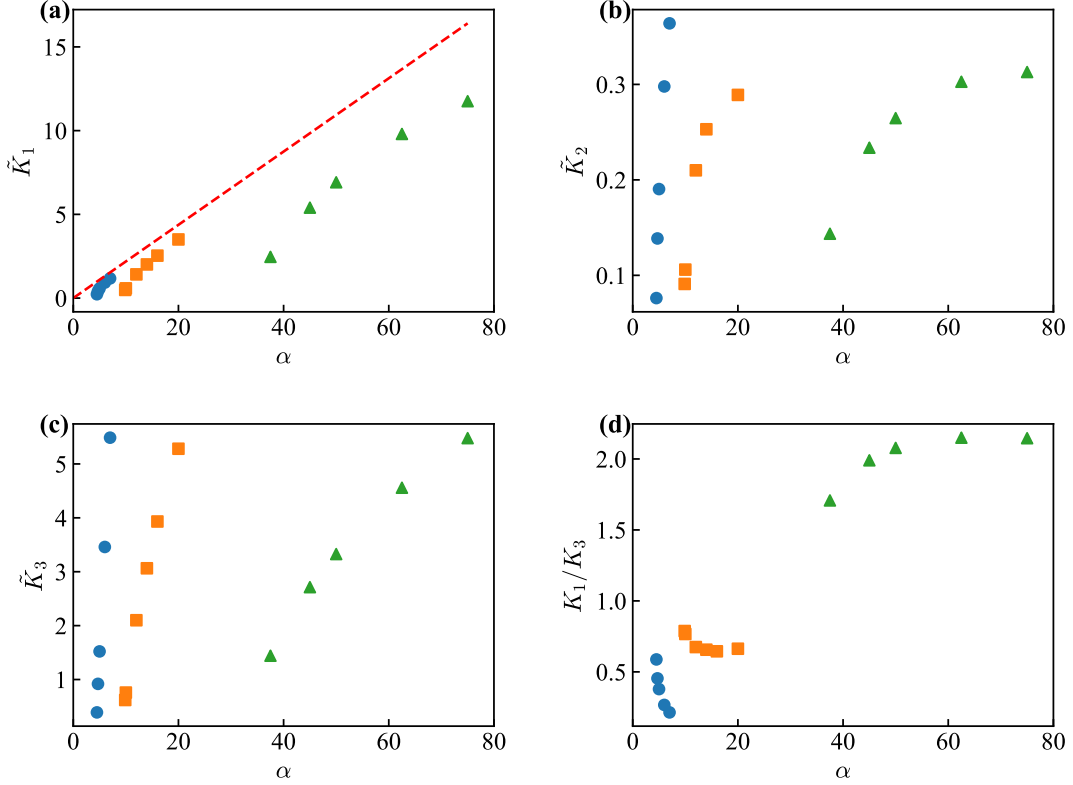


FIG. 2. Scaled Frank elastic constants as functions of α for different values of L/l_p : (a) $\tilde{K}_1$, (b) $\tilde{K}_2$, (c) $\tilde{K}_3$, and (d) the ratio K_1/K_3 . The blue circles, orange squares, and green triangles correspond to rigid chains ($L/l_p = 0$), semiflexible chains with $L/l_p = 1$, and $L/l_p = 5$, respectively. In panel (a), the dashed line indicates the asymptotic scaling in the Onsager limit for strongly aligned rigid rods.

larger than $\tilde{K}_2$, exceeding it by approximately one order of magnitude. $\tilde{K}_2$ and $\tilde{K}_3$ depend strongly on L/l_p : both increase rapidly with α in the rigid-chain limit, but are significantly suppressed as chain flexibility increases. In contrast, $\tilde{K}_1$ exhibits a weaker dependence on chain flexibility and continues to increase with α in a qualitatively similar manner across all values of L/l_p considered. As a consequence, since $\tilde{K}_1$ depends only weakly on L/l_p while $\tilde{K}_3$ is strongly affected, a crossover between $\tilde{K}_1$ and $\tilde{K}_3$ emerges upon increasing chain flexibility, leading to a reversal of the relative magnitude of $\tilde{K}_1$ and $\tilde{K}_3$, consistent with previous studies [22]. To further quantify the anisotropy of the elasticity, we examine the ratio K_1/K_3 as a function of α , as shown in Fig. 2(d). For rigid and moderately flexible chains ($L/l_p = 0$ and 1), we find that $K_1/K_3 < 1$ and decreases with increasing α . In contrast, for more flexible chains ($L/l_p = 5$), $K_1/K_3 > 1$ and increases with α . Overall, the disparity between $\tilde{K}_1$ and $\tilde{K}_3$ becomes more pronounced as α increases.

In order to further assess the strong sensitivity of $\tilde{K}_2$ and $\tilde{K}_3$ on chain flexibility, and the comparatively weak dependence of $\tilde{K}_1$, we next examine the dependence of $\tilde{K}_i$ on chain length L by varying L at fixed l_p , while keeping the chain length density nL/V fixed. It enables direct comparison with experiments where volume fraction is

held fixed. The results are shown in Fig. 3. We find that $\tilde{K}_1$ increases monotonically with L over the range considered. In contrast, $\tilde{K}_2$ and $\tilde{K}_3$ initially increase with L and then approach a plateau at larger L . It indicates that, as chains become more flexible (i.e., increasing L/l_p at fixed l_p), the twist and bend elastic constants become weakly dependent on chain length L and are instead primarily controlled by the persistence length l_p .

To further understand these trends, we compare our results with the analytical prediction in the Onsager limit for strongly aligned rigid rods [15],

$$\beta K_1 = 3\beta K_2 = \frac{7}{32} \frac{1}{D} \left(\frac{n}{V} L^2 D \right), \quad \beta K_3 = \frac{\pi}{48} \frac{1}{D} \left(\frac{n}{V} L^2 D \right)^3, \quad (47)$$

where D is the rod diameter. These results are derived assuming excluded volume interactions between rigid rods, whereas our model employs an explicit MS interaction. Nevertheless, expanding the excluded volume interaction in orientation moments and truncating at second order produces the same quadratic orientational interaction as the MS form, with u_2 in the present model playing a role analogous to the rod diameter D in the Onsager description. Replacing D by u_2 , the asymptotic scaling is consistent with the scaling form obtained in Eq. (46). Within the correspondence $u_2 \sim D$, one has

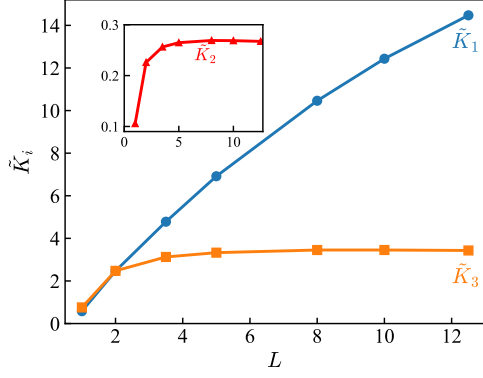


FIG. 3. Scaled Frank elastic constants $\tilde{K}_i$ as functions of chain length L at fixed chain length density $nL/V = 1$ and persistence length $l_p = 1$.

$\alpha \sim \Phi(L/D)$, where Φ is the volume fraction. For fixed L/l_p and L/D , varying α is therefore equivalent, at the scaling level, to scanning the volume fraction. Accordingly, in the high- S asymptotic regime, Eq. (47) predicts $\tilde{K}_1, \tilde{K}_2 \sim \alpha$, while $\tilde{K}_3 \sim \alpha^3$. The relation $\tilde{K}_1 = 3\tilde{K}_2$, which holds exactly in the theory for all S , is approximately reproduced by our numerical results, with better agreement at smaller S and larger deviations emerging at higher S . The discrepancy may arise from the finite truncation of the real spherical harmonics expansion used in the numerical calculations. We also show the asymptotic prediction for $\tilde{K}_1$ from Eq. (47) in Fig. 2(a). Although our numerical results correspond to the intermediate ordering range $S_N \sim 0.4$ – 0.7 and are obtained with MS interaction, the asymptotic result still provides a useful upper reference bound. For $\tilde{K}_3$, the rigid-chain results increase more rapidly than $\tilde{K}_1$ and $\tilde{K}_2$, qualitatively consistent with the stronger growth predicted by Eq. (47), although the observed dependence is weaker than cubic, likely due to finite order spherical harmonics truncation and the distinct asymptotic behavior of the present MS model.

The discussion now turns to the semiflexible regime. As already discussed by Meyer [34], $\tilde{K}_1$ is dominated by the density of chain ends: As the chain length increases, the density of chain ends decreases, making splay deformations more costly. Consequently, $\tilde{K}_1$ depends only weakly on chain flexibility and continues to scale approximately as $\tilde{K}_1 \sim \alpha$. In contrast, in the strongly ordered limit ($S_N \rightarrow 1$) and for very flexible chains ($L/l_p \gg 1$), scaling arguments based on the deflection length picture suggest that $\tilde{K}_2$ and $\tilde{K}_3$ become strongly affected by chain flexibility, yielding [15]

$$\beta K_2 \sim \frac{1}{D} \left(\frac{n}{V} L^2 D \frac{l_p}{L} \right)^{1/3}, \quad \beta K_3 \sim \frac{1}{D} \left(\frac{n}{V} L^2 D \frac{l_p}{L} \right). \quad (48)$$

Using the correspondence $u_2 \sim D$, these relations can be expressed in terms of α as $\tilde{K}_2 \sim (\alpha l_p/L)^{1/3}$ and

$\tilde{K}_3 \sim \alpha l_p/L$. As the chains become more flexible, the ratio l_p/L decreases, suppressing the growth of $\tilde{K}_2$ and $\tilde{K}_3$ with increasing α , consistent with our numerical results. Furthermore, expressing the scaling in terms of the volume fraction Φ , one obtains $\tilde{K}_2 \sim (\Phi l_p/D)^{1/3}$ and $\tilde{K}_3 \sim \Phi l_p/D$. These relations suggest that, for sufficiently flexible chains at fixed Φ , both $\tilde{K}_2$ and $\tilde{K}_3$ are governed primarily by l_p rather than L , consistent with the crossover observed in Fig. 3 near $L/l_p \gtrsim 5$.

We note that Frank elastic constants for wormlike chain models have also been studied using an alternative approach based on a fluctuation free energy [22]. In that work, the MS interaction strength was used as the primary control parameter. Through the scaling form in Eq. 46, those results can be related to the present formulation.

B. Density–nematic scalar order parameter coupling

We next examine the coupled linear response of ρ and Q in both isotropic and nematic states. Numerically, the response is probed by using single mode external fields $h_\rho(x)$ and $h_Q(x)$ with sinusoidal modulation along the x direction. In the isotropic phase, no preferred director exists in the absence of external fields. We find that $h_\rho(x)$ induces nematic alignment with the director parallel to the wavevector (x -direction). Accordingly, for the field coupled to Q , the easy axis $\mathbf{e}$ is chosen parallel to the wavevector. In contrast, the uniform nematic state possesses a spontaneously chosen director $\mathbf{n}$, leading to an anisotropic response that depends on the orientation of $\mathbf{k}$ relative to $\mathbf{n}$. Accordingly, the gradient coefficients are decomposed into parallel and perpendicular components as in Eq. (41). We therefore consider the representative cases $\mathbf{k} \parallel \mathbf{n}$ and $\mathbf{k} \perp \mathbf{n}$.

For both the isotropic and nematic reference states, the extracted $\chi(\mathbf{k})$ is found to be symmetric within numerical accuracy, consistent with the expected symmetry of its inverse $\mathbf{M}(\mathbf{k}) = \chi^{-1}(\mathbf{k})$. The wavevector dependences of the matrix elements of $\mathbf{M}(\mathbf{k})$ are shown in Fig. 4. The matrix elements of $\mathbf{M}(\mathbf{k})$ exhibit a clear quadratic dependence on wavevector components, confirming the validity of the small- k expansion in Eq. (41).

In the isotropic reference state, $M_{\rho Q}$ extrapolates to zero in the $k \rightarrow 0$ limit, consistent with $A_{\rho Q} = 0$ at $S = 0$ in Eq. (39). The gradient coupling nevertheless remains finite, with $K_{\rho Q} > 0$, indicating that spatial variations of ρ and Q prefer out of phase modulations. Correspondingly, higher density regions are associated with $S < 0$ (oblate order), while lower density regions correspond to $S > 0$ (prolate order).

In the nematic reference state, the off-diagonal elements of $M(\mathbf{k})$ are comparable in magnitude to the diagonal ones, indicating strong density–nematic degree of order coupling. Moreover, $A_{\rho Q}$ and $K_{\rho Q}$ are both negative, implying that the system energetically favors in

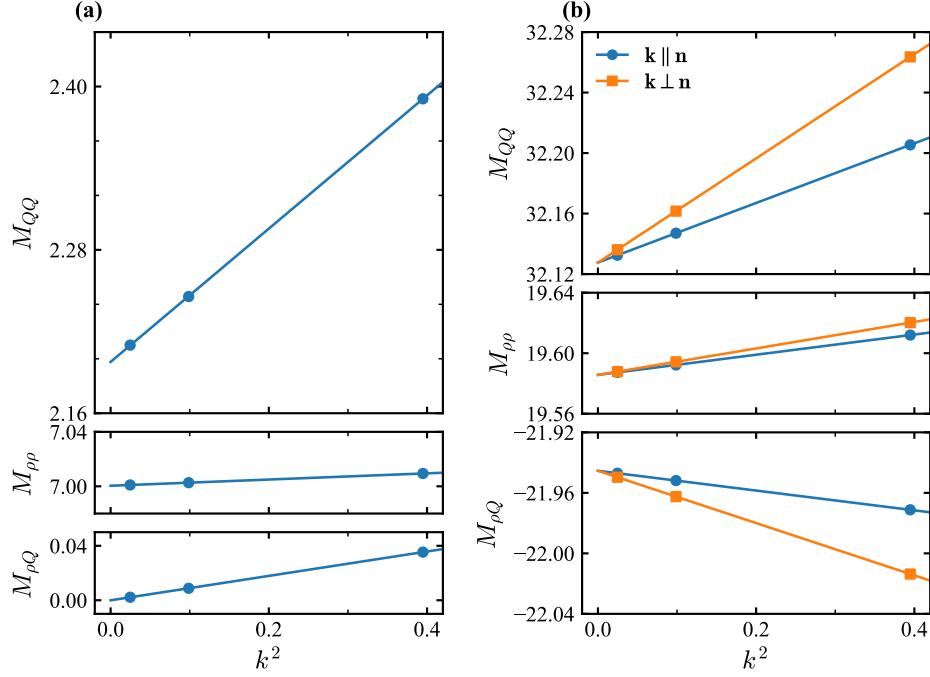


FIG. 4. Elements of the inverse susceptibility matrix M_{ij} as functions of k^2 for (a) the isotropic reference state ($L = l_p = 1$, $n/V = 1$, $u_0 = 6$, $u_2 = 10$) and (b) the nematic reference state ($L = l_p = 1$, $n/V = 1.3$, $u_0 = 10$, $u_2 = 10$). Symbols denote numerical results, while solid lines represent linear fits. (a) The linear fits yield $A_{\rho\rho} \approx 7.00$, $A_{\rho Q} \approx -3.94 \times 10^{-5}$, $A_{QQ} \approx 2.197$, $K_{\rho\rho} \approx 0.0228$, $K_{\rho Q} \approx 0.0896$, and $K_{QQ} \approx 0.4902$. (b) Linear fits for the two directions yield nearly identical intercepts, with $A_{\rho\rho} \approx 19.586$, $A_{\rho Q} \approx -21.945$, and $A_{QQ} \approx 32.128$. In contrast, K_{ij} exhibit clear anisotropy: $K_{\rho\rho}^\perp \approx 0.0875$ and $K_{\rho\rho}^\parallel \approx 0.0667$, $K_{\rho Q}^\perp \approx -0.173$ and $K_{\rho Q}^\parallel \approx -0.0653$, and $K_{QQ}^\perp \approx 0.344$ and $K_{QQ}^\parallel \approx 0.197$.

phase variations of ρ and Q . The results in Fig. 4 are also consistent with the decomposition in Eq. (41), with anisotropic gradient coefficients $K_{ij}^\parallel$ and $K_{ij}^\perp$, while A_{ij} remain direction independent. Neglecting the coupling between ρ and Q , the relation $K_{QQ,\rho\rho}^\parallel < K_{QQ,\rho\rho}^\perp$ would suggest a shorter correlation length for $\mathbf{k} \parallel \mathbf{n}$. However, the coupling renormalizes the effective coefficients $K_{ij}^{\parallel,\perp}$, leading instead to $\xi_Q^\parallel > \xi_Q^\perp$. This trend is consistent with previous experimental and theoretical studies on isotropic-nematic interfaces [8, 20, 21, 35], where the interfacial width is larger for homeotropic alignment (with gradients parallel to the director) than for planar alignment (with gradients perpendicular to the director). This anisotropy is also consistent with the anisotropic core structure commonly observed for $+1/2$ defects [8, 36], where larger core sizes occur along directions corresponding to gradients parallel to the local director.

We now investigate how the coefficients A_{ij} and K_{ij} depend on the parameters u_0 , u_2 , n/V , and L with $L/l_p = 1$ fixed. First, consider the isotropic phase ($S = 0$), the expressions in Eq. (39) predict that A_{ij} depends on chain length density and chain length through the combination nL^2/V . In particular,

$$A_{\rho\rho} = u_0 + (nL^2/V)^{-1}, \quad (S = 0), \quad (49)$$

while the density-nematic order coupling vanishes, $A_{\rho Q} = 0$. Similarly,

$$A_{QQ} = -\frac{9}{4}u_2 + \frac{9}{4}(nL^2/V)^{-1}g'', \quad (S = 0). \quad (50)$$

The numerical results shown in Table I are fully consistent with these predictions. In particular, $A_{\rho Q}$ remains negligibly small within numerical accuracy. From the numerical data, we estimate $g''(S = 0; L/l_p = 1) \approx 11$. The gradient coefficients K_{ij} are essentially insensitive to u_0 and u_2 , and remain nearly unchanged when L is varied at fixed L/l_p . By contrast, reducing the chain density n/V increases all K_{ij} approximately in proportion to $(n/V)^{-1}$.

The situation becomes more complex in the nematic phase because the equilibrium order parameter itself depends on the model parameters through $S_N(\alpha, L/l_p)$. As a result, the coefficients acquire both explicit parameter dependence and implicit dependence through the equilibrium value of S_N , and the associated $g''(S_N; L/l_p)$. Using

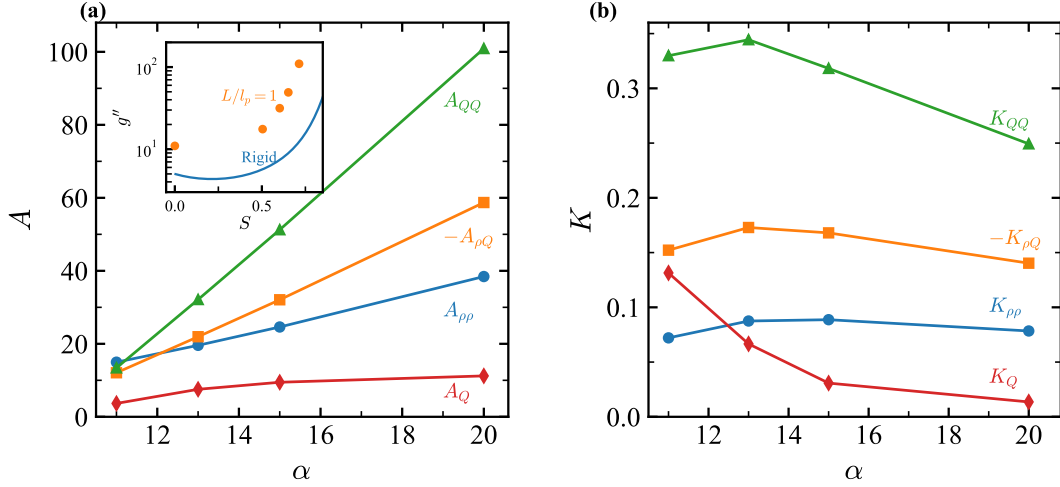


FIG. 5. Dependence of A_{ij} and K_{ij} on α for $L = l_p = 1$, $u_0 = 10$, $u_2 = 10$ and $\mathbf{k} \perp \mathbf{n}$, with α varied through n/V . (a) $A_{\rho\rho}$, $-A_{\rho Q}$, A_{QQ} , and A_Q versus α . Inset: comparison of $g''(S)$ for the rigid-chain limit and the finite-stiffness case $L/l_p = 1$ as a function of S , where the $L/l_p = 1$ data are extracted from A_{ij} . (b) $K_{\rho\rho}$, $-K_{\rho Q}$, K_{QQ} , and K_Q versus α .

TABLE I. Linear response coefficients in the isotropic reference state under different parameter values. Default parameters: $L = l_p = 1$, $n/V = 1$, $u_0 = 6$, $u_2 = 10$.

Case	α	$A_{\rho\rho}$	$K_{\rho\rho}$	$A_{\rho Q} \times 10^5$	$K_{\rho Q}$	A_{QQ}	K_{QQ}
Default	10	7.00	0.0228	-3.94	0.0896	2.197	0.4902
$n/V = 0.5$	5	8.00	0.0456	-7.86	0.1793	26.90	0.9803
$n/V = 0.25$	2.5	10.0	0.0912	-16.6	0.3586	76.30	1.9605
$u_2 = 5$	5	7.00	0.0228	-3.93	0.0896	13.45	0.4901
$L = l_p = 0.5$	2.5	10.0	0.0224	-0.797	0.0879	76.29	0.4818
$u_0 = 5.7$	10	6.70	0.0228	-3.38	0.0896	2.197	0.4902

$1/(L\rho) = u_2/\alpha$, Eq. (39) can be rewritten as

$$\begin{aligned}
 A_{\rho\rho} &= u_0 + u_2 \left[\frac{1}{\alpha} + \frac{S^2}{\alpha} g'' \right], \\
 A_{\rho Q} &= -u_2 \left[\frac{3S}{2\alpha} g'' \right], \\
 A_{QQ} &= u_2 \left[-\frac{9}{4} + \frac{9}{4\alpha} g'' \right].
 \end{aligned} \tag{51}$$

Since $S = S_N(\alpha, L/l_p)$ and $g'' = g''(S_N; L/l_p)$, all terms inside the brackets are functions only of α and L/l_p . This scaling provides a simple interpretation of the large α , rigid-chain limit. In this regime, $g'(S) \sim (1 - S)^{-1}$ near $S = 1$. Combining this asymptotic form with the stationary condition $g'(S_N) = \alpha S_N$ gives $(1 - S_N)^{-1} \sim \alpha$, and therefore $g''(S_N) \sim (1 - S_N)^{-2} \sim \alpha^2$. As a result, all A_{ij} exhibit the same leading asymptotic scaling, $A_{ij} \sim u_2 \alpha$.

We now turn to numerical results for finite chain stiffness in the nematic phase using $L/l_p = 1$ and the $\mathbf{k} \perp \mathbf{n}$ geometry as a representative example, with α varied through n/V . Figure 5(a) shows A_{ij} as functions of α . In the explored range corresponding to $S_N \simeq 0.5$ – 0.7 , the

coefficients A_{ij} display an approximately linear dependence on α . Using Eq. (51), the quantity $g''(S)$ can be extracted independently from each element of A_{ij} . The resulting values from different elements are found to be consistent within numerical accuracy and are shown in the inset of Fig. 5(a), together with the rigid-chain result for comparison. The finite stiffness case ($L/l_p = 1$) exhibits systematically larger values of g'' at the same S , indicating a stronger entropic penalty for nematic ordering. Physically, semiflexible chains possess many local orientational degrees of freedom absent in rigid chains. Increasing nematic order suppresses orientational fluctuations along the chain contour, leading to a substantially larger entropy loss than in the rigid-chain limit. For the gradient coefficients K_{ij} , we perform numerical scans along different paths in parameter space while keeping α and L/l_p fixed. We find that the numerical data are consistent with the scaling form

$$K_{ij} = u_2 L^2 \tilde{K}_{ij}(\alpha, L/l_p), \tag{52}$$

where $\tilde{K}_{ij}$ is a scaling function depending only on α and L/l_p . In contrast to A_{ij} , K_{ij} show only weak dependence on α , as shown in Fig. 5(b). To illustrate the effect of the $\rho - Q$ coupling in the nematic phase, the corresponding effective coefficients are also shown in Fig. 5. As α increases, A_Q initially rises and then saturates, whereas K_Q decreases significantly. Consequently, the correlation length ξ_Q decreases as the system moves deeper into the nematic phase.

We next examine the effect of the excluded volume interaction parameter u_0 . In the isotropic reference state, the coupling term vanishes ($A_{\rho Q} = 0$), so varying u_0 only modifies coefficient $A_{\rho\rho}$ and leaves the Q mode unaffected. In contrast, in the nematic reference state, $A_{\rho Q} \neq 0$. As a result, although varying u_0 directly modi-

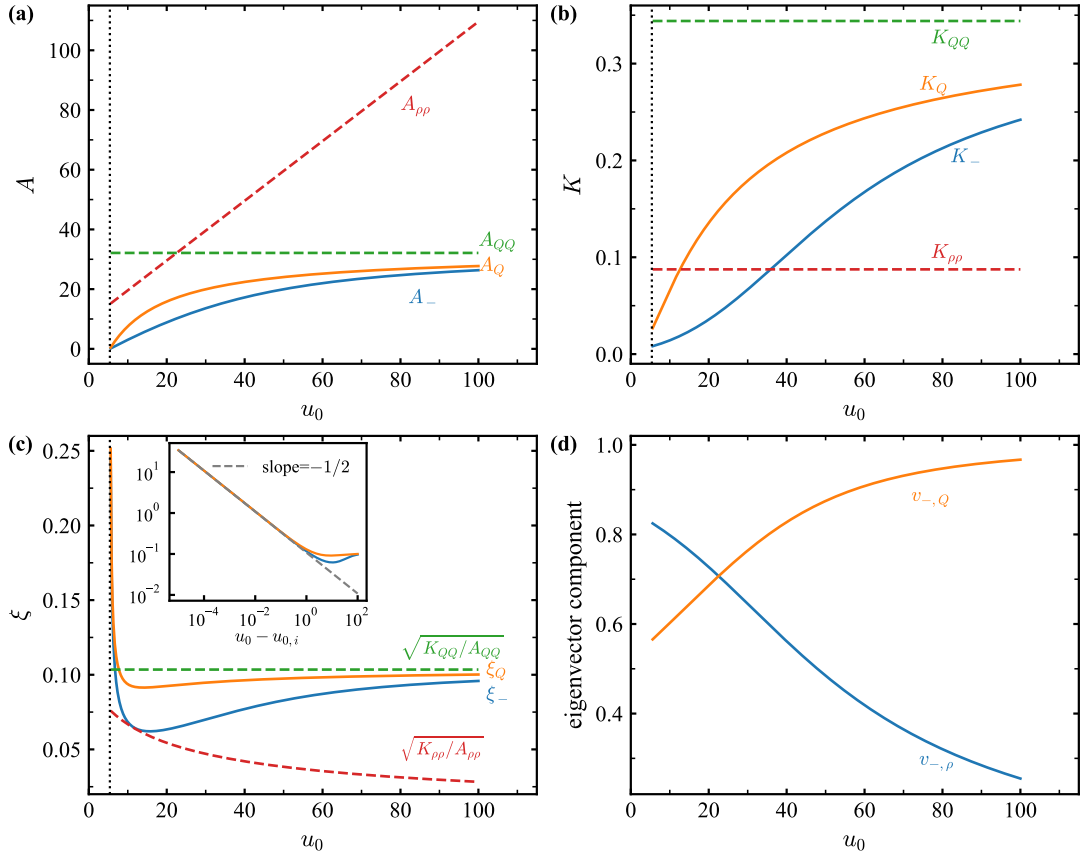


FIG. 6. Dependence on u_0 of (a) the $k = 0$ coefficients, (b) the gradient coefficients, (c) the correlation lengths, and (d) the soft mode eigenvector components. The coefficients are constructed using the A_{ij} and K_{ij} values extracted from the $\mathbf{k} \perp \mathbf{n}$ nematic reference state in Fig. 4, where only $A_{\rho\rho}$ depends on u_0 , through $A_{\rho\rho} = u_0 + 9.586$. The vertical dashed line in (a, b, c) denotes instability point $u_0 = u_{0,i}$, at which $A_Q = A_- = 0$. In panel (c), the inset shows a log-log plot of ξ versus $u_0 - u_{0,i}$, showing the scaling $\xi \sim (u_0 - u_{0,i})^{-1/2}$ near the instability point.

fies only $A_{\rho\rho}$ and leaves S_N unchanged, it can still destabilize the uniform nematic state by softening a coupled $\rho - Q$ mode. As shown in Fig. 6, the $\rho - Q$ coupling reduces both A_- and A_Q relative to $A_{\rho\rho}$ and A_{QQ} . As u_0 increases, both A_- and A_Q approach A_{QQ} , while the corresponding gradient coefficients approach K_{QQ} , reflecting the suppression of density fluctuations and the evolution of the soft mode toward a pure Q mode. Conversely, both A_- and A_Q vanish at a threshold value $u_{0,i}$, corresponding to the nematic spinodal identified in Sec. III, whereas the corresponding gradient coefficients remain finite, leading to the divergence of the correlation lengths shown in Fig. 6(c). The inset of Fig. 6(c) shows that the two correlation lengths ξ_- and ξ_Q follow the same scaling $\xi \sim (u_0 - u_{0,i})^{-1/2}$, reflecting the fact that A_- and A_Q vanish linearly with u_0 near the spinodal. We next examine the $k = 0$ soft mode eigenvector, $\mathbf{v}_- = (v_{-,\rho}, v_{-,Q})$, shown in Fig. 6(d). The positive ratio $v_{-,\rho}/v_{-,Q}$ indicates positively correlated ρ and Q variations. As u_0 increases, the ρ component is progressively suppressed, causing the soft mode to become increasingly dominated by nematic order variations.

C. Topological defects: Core size and elastic anisotropy

This section describes our results concerning director profiles induced by disclinations in the medium, and the effects due to Frank elastic constant anisotropy and the magnitude of correlation lengths. In particular, we focus on how defect core size and its anisotropy in the density weighted order parameter profile ρS are determined by the correlation lengths, and how the director configuration is modified by elastic anisotropy.

Two representative $+1/2$ defects which exhibit markedly different elastic anisotropies are considered: one for rigid chains ($l_p \rightarrow \infty$) and the other for semiflexible chains with $L/l_p = 5$. The SCFT equations are solved numerically as described in Appendix B, and the corresponding ρS configurations are shown in Fig. 7(a, d). For the rigid case, the core anisotropy is relatively weak, whereas for $L/l_p = 5$ it is much more pronounced: the core radius on the side where the gradient is parallel to $\mathbf{n}$, denoted $r_{\parallel}$, is significantly larger than that on the side where the gradient is perpendicular to $\mathbf{n}$, denoted

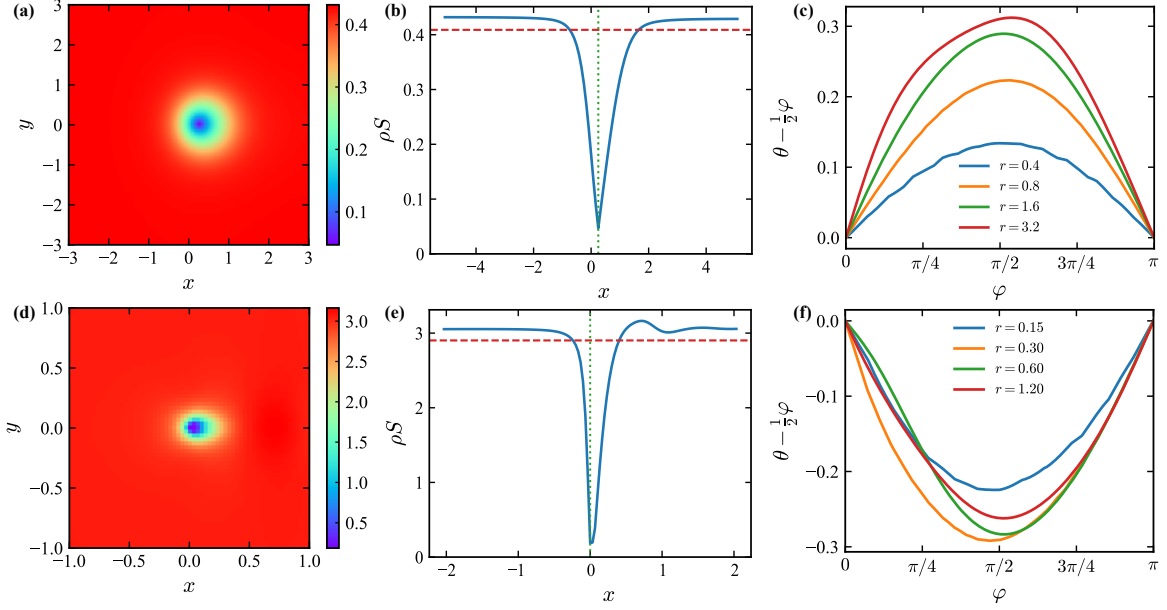


FIG. 7. Comparison of defect structures in the rigid and semiflexible cases. Top row: rigid case with $L = 1$, $l_p \rightarrow \infty$ and $n/V = 0.6$. Bottom row: semiflexible case with $L = 1$, $l_p = 0.2$, and $n/V = 5$. Both cases use $u_0 = u_2 = 10$. From left to right, each row shows the ρS field, the corresponding ρS profile along the x direction through the defect core, and the director angle deviation from the isotropic configuration at different distances to the core center. In panels (b) and (e), the green dotted lines indicate the defect core center, defined by the minimum of ρS , while the red dashed lines denote 95% of the bulk ρS value used to determine the core radius.

$r_\perp$. In practice, we estimate the core radius from the numerical profiles by defining it as the distance from the defect center (identified as the location where ρS attains its minimum) to the point at which ρS recovers to 95% of its bulk value, as shown in Fig. 7(b, e). Taking the defect center as the origin, we introduce polar coordinates (r, φ) , where φ is the polar angle and r is the radial distance. We denote by $\theta(r, \varphi)$ the director angle at position (r, φ) , measured relative to the $\varphi = 0$ axis. We use the same notation θ as in the previous section, the meaning should be clear from the context. For isotropic elasticity, the director field is given by $\theta(r, \varphi) = \varphi/2$. For a $+1/2$ defect in an elastically anisotropic medium, a perturbative expansion to the Dzyaloshinskii solution [8, 37, 38] in small anisotropy gives,

$$\theta(r, \varphi) = \frac{\varphi}{2} + \frac{3}{4}\epsilon \sin \varphi + \mathcal{O}(\epsilon^2), \quad (53)$$

where the elastic anisotropy parameter $\epsilon = (K_3 - K_1)/(K_1 + K_3)$. Strictly speaking, ϵ should depend on position, $\epsilon = \epsilon(r, \varphi)$, because both K_1 and K_3 vary with the local density and nematic order. Therefore, the first order correction is expected to depend on r , and may also deviate from a simple sinusoidal form if ϵ itself acquires angular dependence.

For the rigid chain system, we obtain $\epsilon \approx 0.5$ from the linear response analysis of the uniform nematic state under the same parameters, while for $L/l_p = 5$ we find $\epsilon \approx -0.36$. In Fig. 7(c, f), we plot the equilibrium $\theta - \varphi/2$ at different radii. The results show that for both the rigid

chain system and the case $L/l_p = 5$, the angular deviation $\theta - \varphi/2$ is well approximated by a sinusoidal form. The amplitude of the deviation generally decreases as one approaches the defect center. This behavior is consistent with the reduction of elastic anisotropy near the core, where ρ and S are lowered, as reflected in Fig. 2(d). For the case of $L/l_p = 5$, the amplitude does not increase monotonically with distance from the core. It is likely influenced by the choice of initial conditions and square boundary, which can affect the director configuration away from the core.

An important question concerns the characteristic size of a defect core, and its relation to the equilibrium correlation length. In the case of lyotropic chromonics, core sizes are anomalously large [8], which we believe may be caused by aggregate scission and related density changes in regions of large director gradients. This anomalous size is not observed in our current calculations which only allow chain conformation fluctuations and the associated local density changes, but no chain polydispersity. We compare the core size extracted from the profile of ρS , taking $r_\perp$ as a representative measure, with the corresponding correlation length ξ_Q . As shown in Fig. 8, the data obtained from scans over n/V , L/l_p , and u_0 display approximate proportionality between the core size and ξ_Q , although with slightly different slopes for the different parameter scans. This indicates that the characteristic defect core size is largely determined by the nematic equilibrium correlation length. Although linear response analysis predicts a divergence of ξ_Q at the nematic spin-

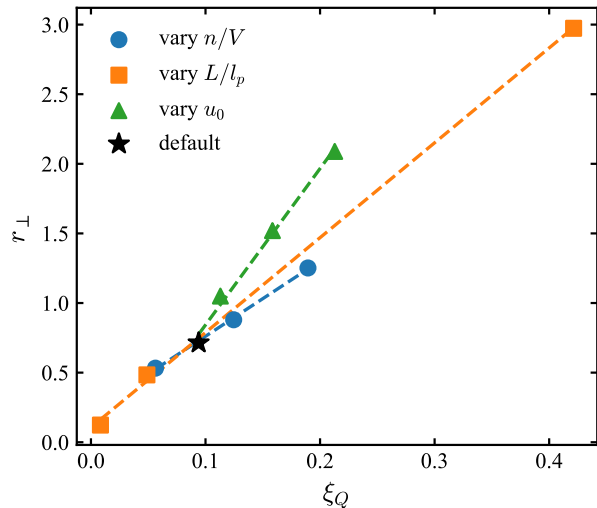


FIG. 8. Core radius $r_\perp$ as a function of the correlation length ξ_Q for parameter scans over n/V , L/l_p , and u_0 . Circles, squares, and triangles denote the three parameter scans, respectively, while the dashed lines are linear fits, highlighting the approximately proportional relation between $r_\perp$ and ξ_Q . Increasing either u_0 or n/V reduces ξ_Q . In the L/l_p scan (with $L = 1$ fixed), increasing l_p leads to a larger ξ_Q . The default parameters are $u_0 = u_2 = 10$, $n/V = 1.3$, and $L = l_p = 1$. The u_0 scan uses $u_0 = 5.7, 6$, and 7 . The chain number density scan uses $n/V = 1.1, 1.2$, and 1.5 . For the chain stiffness scan, $L = 1$ is fixed and $l_p = 0.1$ ($n/V = 9.2$), 0.5 ($n/V = 2.2$), and ∞ ($n/V = 0.5$) are chosen, with n/V adjusted to maintain $S_N \approx 0.6$.

odal, this divergence is preempted by the appearance of an isotropic domain (tactoid) at the core in our defect configurations. As the defect core locally suppresses nematic order, phase separation occurs before the spinodal is reached, effectively cutting off the asymptotic growth of ξ_Q .

VI. CONCLUSION

We have investigated elasticity, density–nematic order coupling, correlation lengths, and disclination structure in a semiflexible nematic within a wormlike chain SCFT framework. The aligning interaction between chain segments has been assumed to be of the Maier–Saupe form. A solvent has been introduced implicitly by adding to the Hamiltonian an isotropic excluded volume interaction term between chains, which depends on the Flory–Huggins enthalpy of interaction between chain and solvent molecules.

Concerning uniform equilibrium states, an effective free energy has been obtained in terms of the chain segment density ρ , and the uniaxial nematic order parameter S , revealing the coupling between the two. In the rigid-chain limit, the reduction in conformational entropy per chain $g(S)$ due to nematic order exhibits a logarithmic

singularity as $S \rightarrow 1$ (perfect order). The resulting free energy provides a description of isotropic–nematic coexistence and spinodal instabilities. In particular, the binodal and spinodal boundaries can be determined directly from the uniform free energy or, equivalently, from the grand potential formulation. Unlike the conventional Landau–de Gennes theory, which is based on a low-order expansion about $S = 0$, the present free energy retains the physical bounds $-1/2 \leq S \leq 1$ and the associated singular behavior at the boundaries.

Using SCFT numerical calculations to perform a linear response analysis, we obtain Frank elastic constants as a function of density, as well as coupled susceptibilities for density and nematic order fluctuations. While director distortions are found to remain effectively decoupled from density and scalar order parameter variations at linear order, the density and the uniaxial nematic order parameter exhibit strong coupling in the nematic phase. The elastic constants follow a scaling form $\beta K_i = u_2^{-1} \tilde{K}_i(\alpha, L/l_p)$, where $\alpha = nL^2u_2/V$ is an effective strength of the alignment interaction, and L/l_p is the ratio between chain and persistence lengths. For rigid chains, the numerical results are qualitatively consistent with the asymptotic scaling behavior in the Onsager model. In the semiflexible regime, bend and twist elastic constants are strongly reduced by chain flexibility, while splay elasticity depends comparatively weakly on L/l_p . As a result, increasing chain flexibility produces a crossover from bend-dominated to splay-dominated elasticity, signaled by a reversal of the ratio K_1/K_3 .

The coupled density–nematic order response further reveals strong renormalization effects. Although the diagonal gradient terms of the inverse susceptibility matrix satisfy $K_{QQ}^\parallel < K_{QQ}^\perp$, the coupling to density modifies the effective response and produces the opposite anisotropy for correlation lengths, $\xi_Q^\parallel > \xi_Q^\perp$. In solution-based lyotropic nematics, the excluded volume parameter u_0 reflects solvent mediated interactions and osmotic compressibility effects, which change with solvent conditions, temperature, or ionic strength. Increasing u_0 reduces density variations, and drives the system toward an effectively incompressible nematic solely dominated by nematic order fluctuations. For compressible systems, the calculations reveal a nematic spinodal arising from the coupling between density fluctuations and the degree of nematic order. As the spinodal is approached, the associated correlation lengths diverge as $\xi \sim (u_0 - u_{0,i})^{-1/2}$, where $u_{0,i}$ denotes the instability threshold.

SCFT calculations have also been used to study the structure of topological defects in two dimensions, albeit with a three-dimensional tensor order parameter (thin film geometry). The anisotropy of the director profile around a defect core is consistent with the anisotropy of the elastic constants obtained from the linear response analysis, and the classical Dzyaloshinskii solution for a uniaxial nematic. The characteristic core size obtained from SCFT calculations scales approximately linearly with the correlation length ξ_Q , in agreement with

Landau–de Gennes predicted scaling. Director field distortions due to elastic anisotropy increase significantly in the semiflexible regime.

Overall, the present work provides a microscopic framework linking molecular flexibility, density–nematic order coupling, elastic anisotropy, and defect structure in semiflexible nematics. The results demonstrate how collective continuum properties can emerge from microscopic chain statistics within a unified SCFT description. Several directions remain open for future work. In particular, it would be valuable to derive Landau–de Gennes type gradient free energy directly from the SCFT framework, thereby establishing a more systematic connection between microscopic chain statistics and phenomenological continuum theories. The present results also suggest possible extensions to the Landau–de Gennes theory to incorporate density variations, particularly in semiflexible and lyotropic nematic systems where density–nematic order coupling plays an important role. More generally, the framework developed here provides a basis for future extensions to self-assembled, polydisperse chains featuring reversible aggregation.

ACKNOWLEDGMENTS

This research has been supported by the National Science Foundation under contract DMR-2223707, by the Minnesota Supercomputing Institute of the University of Minnesota, and by the Advanced Cyberinfrastructure Coordination Ecosystem: Services & Support (ACCESS) program, which is supported by U.S. National Science Foundation grants 2138259, 2138286, 2138307, 2137603, and 2138296.

Appendix A: Grand potential formulation

The grand potential density is defined as

$$\beta\omega(\mu, S) = \beta f_{\text{eff}}(\rho, S) - \frac{\beta\mu}{L}\rho, \quad (\text{A1})$$

with the density ρ determined implicitly by $\partial(\beta f_{\text{eff}})/\partial\rho = \beta\mu/L$. The osmotic pressure $\beta P = \rho\partial(\beta f_{\text{eff}})/\partial\rho - \beta f_{\text{eff}}$ implies $\beta\omega = -\beta P$. Therefore, equality of osmotic pressure is equivalent to equality of the grand potential density, establishing the equivalence between the coexistence conditions.

At fixed μ , the curvature with respect to S is

$$\left(\frac{\partial^2\beta\omega}{\partial S^2}\right)_\mu = A_{SS} - \frac{A_{\rho S}^2}{A_{\rho\rho}}, \quad (\text{A2})$$

where A_{ij} are the Hessian matrix elements defined in the main text. Since $A_{\rho\rho} > 0$ throughout the physically relevant parameter range, the spinodal condition $\partial^2(\beta\omega)/\partial S^2 = 0$ is equivalent to $\det A = 0$. Thus, the spinodal obtained from the vanishing curvature of the

grand potential density at fixed chemical potential is identical to the spinodal determined from the Hessian stability criterion.

Appendix B: Numerical method for SCFT

The SCFT equations, Eqs. (7) and (8), are solved numerically using a real spherical-harmonic expansion together with an iterative procedure, following the numerical methods developed in our previous work [20] and in Ref. [25]. The Python code used in this work is available at Zenodo [39].

An orientation-dependent field is expanded in real spherical harmonics as

$$f(\mathbf{r}, \mathbf{u}) = \sum_{l,m} f_l^m(\mathbf{r}) \tilde{Y}_l^m(\mathbf{u}), \quad (\text{B1})$$

where f represents fields such as the propagator $q(\mathbf{r}, \mathbf{u}, s)$, density field $\rho(\mathbf{r}, \mathbf{u})$, and auxiliary field $w(\mathbf{r}, \mathbf{u})$, with the expansion for the propagator understood to be performed at fixed contour variable s . In the numerical implementation, the spherical-harmonic expansion is truncated at $l = 4$, and the spatial domain is discretized on a Cartesian grid. For linear response calculations, spatial variations are restricted to one dimension, such that all fields depend only on x , i.e., $f_l^m = f_l^m(x_i)$, with the y and z directions assumed to be uniform. For defect calculations, spatial variations are resolved in two dimensions, and fields are represented on a grid (x_i, y_j) , while the z direction remains uniform, i.e., $f_l^m = f_l^m(x_i, y_j)$. Although spatial variations are restricted to one or two dimensions, the orientational degrees of freedom remain fully three-dimensional through the real spherical-harmonic expansion. The corresponding tensor order parameter $\mathbf{Q}$ is therefore defined in three dimensions, although certain components may vanish due to the symmetry of the imposed geometry.

The contour variable is defined over the interval $s \in [0, L]$, and the propagator $q_l^m(\mathbf{r}, s)$ is obtained by solving the MDE using a Lax–Wendroff scheme [20, 40] with a uniform contour step size $\Delta s = L/200$. Picard iteration with simple mixing [21, 41] is employed, with the fields updated according to

$$w^{(i+1)} = (1 - \gamma)w^{(i)} + \gamma w_{\text{new}}^{(i)}, \quad (\text{B2})$$

where i labels the iteration step and γ is the mixing parameter. Stable convergence typically requires a relatively small value of γ (normally $\gamma < 0.1$), which leads to slow convergence. To improve convergence efficiency while maintaining stability, a periodically enhanced mixing strategy is adopted, in which a small baseline value of γ is used for most iterations, while a larger value is temporarily applied for several iterations before reverting to the baseline value.

For linear response calculations, the external fields are imposed with a sinusoidal modulation along the x direction. The computations are performed for wavelengths

$\lambda \sim 10\text{--}40L$, corresponding to the small- k regime. In the numerical implementation, the system size L_x is taken to be equal to the imposed wavelength, and periodic boundary conditions are applied. The domain in the x direction is discretized into 200 uniformly spaced grid points. The amplitude of the prescribed external fields is chosen in the range $h \sim 10^{-6}\text{--}10^{-5}$. We have verified that, within this range, the induced response amplitude scale linearly with the applied field for fixed k , ensuring that all calculations are performed within the linear response regime. In the linear response calculations, convergence can be further accelerated by temporarily adjusting the amplitude of external field during the iteration: when the instantaneous response amplitude remains below the target value, a larger field amplitude is applied to drive the response upward more rapidly, while a smaller field

amplitude is used when the response exceeds the target value. The adjustment range is progressively reduced, and the external field is repeatedly restored to the prescribed value until the response converges to the desired accuracy.

For the topological defect calculations, the system is defined in a two-dimensional square domain with lateral dimensions $L_x = L_y \sim 4\text{--}10L$, discretized into 120×120 uniform grid points. We adopt a similar initialization procedure as in Ref. [20], specifying the initial configuration by a radially varying scalar order parameter and a prescribed director field corresponding to a $+1/2$ disclination. The initial director fields are constructed using the values of ϵ extracted from the linear response analysis. The system is then relaxed to equilibrium through the iterative procedure.

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